

Kai Yan

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Research Interest

My research interest is demonstration for better policy learning. My main areas of interest are: 1) reinforcement learning with demonstrations / imitation learning and 2) LLM / VLM agent / post-training. Apart from that, I have extensive experience in 1) multi-agent reinforcement learning, 2) prediction-and-optimization algorithms, 3) decision transformer, 4) LLM many-shot in-context learning.

Education

University of Illinois Urbana-Champaign

Ph.D. in Computer Science

Urbana, Illinois, US

Sept 2021 - Current

Peking University

B.S. in Computer Science

Beijing, China

Sept 2017 - Jun 2021

- Member of the Turing Class honor program; graduate with *Summa Cum Laude*
- Serve as the vice president of the student union for the school of EECS; organize multiple large events with hundreds of participants

Academic Experience

University of Illinois Urbana-Champaign

Graduate Research Assistant (Co-advised by Prof. Alexander G. Schwing and Prof. Yuxiong Wang)

Urbana, Illinois, US

Sept 2021 - 15th May, 2027 (Estimated)

- Work on demonstration-guided reinforcement learning, imitation learning and Large Language Model reasoning / decision-making
- Extensive experience in normalizing flows, convex optimization, optimal transport, decision transformer, LLM/VLM agents, etc.

Robotics Institute @ Carnegie Mellon University

Visiting Student for Summer Research (Advised by Prof. Changliu Liu)

Remote

Jun 2020 - Jan 2021

- Build a pandemic simulator with reinforcement learning agents as individuals
- Design algorithm for million-level multi-agent training; use Cython for parallelization & acceleration

National Engineering Laboratory for Video Technology @ Peking University

Undergraduate Research Student (Advised by Prof. Zongqing Lu)

Beijing, China

Jun 2019 - Feb 2020

- Work on adversarial agent in multi-agent systems
- Extensively read papers in reinforcement learning, multi-agent systems and algorithmic game theory, and implement RL algorithms from scratch
- Write a survey on algorithmic game theory and reinforcement learning

AILab @ Peking University

Undergraduate Research Student (Advised by Prof. Wenxin Li)

Beijing, China

Nov 2018 - Mar 2019

- Study *Introduction to Reinforcement Learning* and papers on Reinforcement Learning
- Program the judge for the game *Tank2* on Botzone, an AI platform for education and the judge is extensively used for freshmen's course projects

Industry Experience

NVIDIA

Research Intern (Advised by Dr. Boris Ivanovic)

Santa Clara, US

May 2026 - Oct 2026 (Estimated)

- VLA post-training for autonomous driving (prospective)

Apple

Research Intern (Advised by Prof. Philipp Krähenbühl)

Sunnyvale, US

May 2025 - Aug 2025

- Train a VLM spatial reasoning agent with code interpreter for reasoning with SFT and RL

ByteDance Seed

Student Researcher (Advised by Dr. Jiecao Chen)

San Jose, US

Jun 2024 - May 2025

- Build a many-shot in-context learning benchmark and supervised finetuning for LLM self-plan
- Publish two benchmarks on testing LLM's many-shot inductive reasoning ability and reasoning robustness under subtly changed conditions
- Using verl for end-to-end training of a tool-using LLM agent with code interpreter as the environment

Microsoft Research Asia

Research Intern (Co-advised by Dr. Jie Yan and Dr. Chuan Luo)

Beijing, China

Sept 2020 - Jun 2021

- Design an prediction-and-optimization surrogate framework for linear/quadratic optimization problems with $\max(\cdot, 0)$ operator, where the parameterized objective needs to be predicted from data
- Awarded *Stars of Tomorrow* title

Teaching Experience

University of Illinois Urbana-Champaign

Teaching Assistant (Computational Photography)

- Host Office hours every week, grade homework and set up exams

Urbana, Illinois, US

Jan 2024 - May 2024

Peking University

Teaching Assistant (Introduction to Computer Systems)

- Host a 2-hour seminar of 15 students each week, revising classes and introducing cutting-edge applications of the knowledge taught in class
- Teach how to write & organize lecture notes; grade homework every week

Beijing, China

Sept 2020 - Jan 2021

Publications & Preprints

Conference

- Siqi Yang, **Kai Yan**, Alexander G. Schwing and Yuxiong Wang. Latent Wasserstein Adversarial Imitation Learning. In ICLR, 2026.
- **Kai Yan**, Yufei Xu, Zhengyin Du, Xuesong Yao, Zheyu Wang, Xiaowen Guo, Jiecao Chen. Recitation over Reasoning: Can Your Language Models Really Solve Simple Reasoning Problems? In IJCNLP-AACL, 2025.
- **Kai Yan**, Zhan Ling, Kang Liu, Yifan Yang, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, Jiecao Chen. MIR-Bench: Benchmarking LLM's Long-Context Intelligence via Many-Shot In-Context Inductive Reasoning. In NeurIPS Datasets & Benchmarks Track, 2025.
- Anjiang Wei, Tarun Suresh, Jiannan Cao, Naveen Kannan, Yuheng Wu, **Kai Yan**, Thiago S. F. X. Teixeira, Ke Wang, Alex Aiken. CodeARC: Benchmarking Reasoning Capabilities of LLM Agents for Inductive Program Synthesis from Input-Output Examples. In COLM, 2025.
- **Kai Yan**, Alexander G. Schwing and Yuxiong Wang. Reinforcement Learning Gradients as Vitamin for Online Finetuning Decision Transformers. In NeurIPS, 2024. (Spotlight; top 2% submission)
- Andy Zhou, **Kai Yan**, Michal Shlapentokh-Rothman, Haohan Wang, Yuxiong Wang. Language Agent Tree Search Unifies Reasoning Acting and Planning in Language Models, In ICML, 2024. (Github > 800 Stars; integrated in LangChain; invited talk @ Autogen Discord)
- **Kai Yan**, Alexander G. Schwing and Yuxiong Wang. Offline Imitation from Observation via Primal Wasserstein State Occupancy Matching. In ICML, 2024.
- **Kai Yan**, Alexander G. Schwing and Yuxiong Wang. A Simple Solution for Offline Imitation from Observations and Examples with Possibly Incomplete Trajectories. In NeurIPS, 2023.
- **Kai Yan**, Alexander G. Schwing and Yuxiong Wang. CEIP: Combining Explicit and Implicit Priors for Reinforcement Learning with Demonstrations. In NeurIPS, 2022.
- **Kai Yan**, Jie Yan, Chuan Luo, Liting Chen, Qingwei Lin and Dongmei Zhang. A Surrogate Objective Framework for Prediction+Optimization with Soft Constraints. In NeurIPS, 2021.

Preprints

- **Kai Yan**, Alexander G. Schwing and Yuxiong Wang. Boosting Reinforcement Learning with Verifiable Rewards via Randomly Selected Few-Shot Guidance. arXiv:2605.15012, 2026.
- Zhan Ling, Kang Liu, **Kai Yan**, Yifan Yang, Weijian Lin, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, Jiecao Chen. LongReason: A Synthetic Long-Context Reasoning Benchmark via Context Expansion. arXiv:2501.15089, 2025.
- **Kai Yan**¹, Zhenggang Tang¹, Liting Sun, Wei Zhan, Changliu Liu. A Microscopic Pandemic Simulator for Pandemic Prediction Using Scalable Million-Agent Reinforcement Learning. arXiv:2108.06589, 2021.
- **Kai Yan**¹, Yunlong Lu¹. Algorithms in Multi-Agent Systems: A Holistic Perspective from Reinforcement Learning and Game Theory. arXiv:2001.06487, 2020.

¹Equal contribution.

Honors & Awards

Jan. 2025	Graduate College Conference Presentation Award , University of Illinois at Urbana-Champaign	Urbana, Illinois, US
Feb. 2023	Graduate College Conference Presentation Award , University of Illinois at Urbana-Champaign	Urbana, Illinois, US
Jul. 2021	Stars of Tomorrow , Microsoft Research Asia	Beijing, China
Nov. 2020	John Hopcroft Scholarship , Center on Frontiers of Computing Studies (CFCS) @ PKU	Beijing, China
Dec. 2019	National Scholarship & Merit Student , Peking University	Beijing, China
Jan. 2019	Meritorious Winner , International Mathematical Contest in Modeling (MCM)	Remote
Sept. 2018	Founder Scholarship & Merit Student , Peking University	Beijing, China
Apr. 2018	2nd Award , PKU ACM Campus	Beijing, China
Jul. 2016	Silver Medal , National Olympiad in Informatics (NOI)	Mianyang, China
May. 2016	Silver Award , Asian-Pacific Informatics Olympiad (APIO)	Beijing, China

Services

Reviewer

- AAAI Conference on Artificial Intelligence (**AAAI**), **2025**
- Transactions on Machine Learning Research (**TMLR**), **2024-2026** (2025 Expert reviewer certification)
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), **2023-2026**
- Neural Information Processing Systems (**NeurIPS**), **2023-2025** (2024 Top 10% reviewer)
- International Conference on Learning Representations (**ICLR**), **2024-2026**
- International Conference on Machine Learning (**ICML**), **2024-2026** (2026 Gold reviewer)
- European Conference on Computer Vision (**ECCV**), **2024;2026**
- International Conference on Computer Vision (**ICCV**), **2025**
- IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), **2026**

Misc.

- Vice President of the PKU EECS Student Union, 2019-2020

Invited Talks

Oct. 2024	Guest Lecturer on Agentic Foundation Models , University of Michigan	Online
Sep. 2024	Talks on Language Agent Tree Search (LATS) , AutoGen Discord Channel	Online

Skills

Programming & Software	Python (Pytorch for deep learning, CVXPY / Gurobi / Google ORTools for optimization and Cython for high-performance), C/C++, Java, Matlab, HTML/CSS, JavaScript (Node.js), Qt, tex, Microsoft Office, Wireshark
Language	English (Professional with TOEFL 117/120), Chinese (Native)